

A seven-eighths theorem for Hecke characters over arbitrary number fields

Original source results, field transfer, and one closure for each fixed infinity type

AI-assisted research draft*

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Abstract

Relative to the original theta constructions and canonical analytic proofs in OpenAI's quasi-Riemann and primitive-roots papers, every unitary Hecke character over every number field has a zero-free half-plane $\operatorname{Re} s > 7/8$. We prove the necessary transfer to ideal norms, unit quotients, and a fixed arbitrary infinity type. In particular, the general-field marked inverse and paired fourth moments are consequences of the cited constructions and the transfer arguments supplied here, rather than additional hypotheses. The proof passes from F to $K = F(\zeta_{12})$ and works with finite-order twists of one fixed character and its conjugate. A single-row sixth-power amplification gives $11/12$ for this family; the original dynamic fourth-moment capacity then closes directly at $7/8$. The real losses and slot system are fixed before the target conductor, while height orders and constants may depend on it. The long original canonical proofs are retained as source inputs and are not independently re-proved in full. No earlier extension manuscript is used as a theorem source.

1 Main theorem

The two primary sources are OpenAI's original quasi-Riemann paper [1], denoted Q, and its primitive-roots paper [2], denoted P. We distinguish the original analytic work that is cited from the field and infinity-type transfer proved in this paper.

Analytic input 1.1 (Original source results). *Retain the theta construction and local identities of P, Sections 3 and 5–7 and Appendix A, and the physical estimates, zero detection, and canonical moment arguments of Q, Sections 4–18, with the finite-seminorm, strict-width, common-mask, and polynomial-height assertions established there. The full canonical constructions and reductions are used, not just the published zero-free theorems. Their extension to a general field and a fixed unitary infinity type is proved below.*

The usual Hecke functional equation, class field theory, and the number-field quadratic and higher-order large sieves enter through the classical results used in Q and P. No earlier extension manuscript is an additional input.

Theorem 1.2 (Arbitrary number fields). *Under Source Input 1.1, let F be any number field. Every continuous unitary Hecke character $\psi : F^\times \backslash \mathbb{A}_F^\times \rightarrow S^1$ satisfies*

$$\boxed{L_F(s, \psi) \neq 0 \quad (\operatorname{Re} s > 7/8).} \quad (1.1)$$

Finite and infinite order are both allowed, with arbitrary infinity type. A pole for a unitary norm character is permitted. Additional deleted Euler factors do not change the assertion. The

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boundary is independent of the field, conductor, infinity type, and height; implied constants and lower thresholds need not be uniform in these data.

For a prescribed ψ , the proof uses only the fixed-type family (2.2) over $K = F(\zeta_{12})$, with $\omega = \psi \circ N_{K/F}$. It requires no estimate uniform over all infinity types and no additional general-field moment hypothesis.

For a character in the normalization $\Psi(\mathfrak{a}) = \psi(\mathfrak{a})(N\mathfrak{a})^u$, $u \in \mathbb{R}$, one has $L_F(s, \Psi) = L_F(s - u, \psi)$. Thus the corresponding boundary is $u + 7/8$. In particular, an algebraic character with ideal-value absolute size $(N\mathfrak{a})^{w/2}$ has boundary $w/2 + 7/8$ in its unnormalized variable.

There is one analytic closure:

$$\text{marked inverse moment} \implies 11/12 \implies \text{dynamic fourth moment and closure} \implies 7/8.$$

The transfer is established in Section 3 and Appendix A; the descent from K to F is elementary after standard Hecke continuation. The source-relative qualification concerns the original long analytic constructions, not a separate assumption of their desired general-field conclusion.

2 Field reduction and the fixed-type family

2.1 Abelian descent

The extension $K = F(\zeta_{12})$ is abelian over F , has degree at most four, and is totally imaginary. It need not be cyclotomic over $\mathbb{Q}$. The distinction matters: the cyclotomic scope of P, Theorem 1.2, does not by itself cover this K .

We use ideal-value normalization for norm twists: $N_K^{it}(\mathfrak{a}) = (N\mathfrak{a})^{it}$. Such a twist replaces $L_K(s, \eta)$ by $L_K(s - it, \eta)$.

Lemma 2.1 (Descent, including possible poles). *Let K/F be a finite abelian extension and let ψ be a unitary Hecke character of F . If $L_K(s, \psi \circ N_{K/F})$ has no zeros in $\text{Re } s > \sigma_0$, where $\sigma_0 > 0$, then $L_F(s, \psi)$ has no zeros there.*

Proof. Let $G = \text{Gal}(K/F)$, and regard $\widehat{G}$ as a group of finite-order Hecke characters of F by class field theory. Choose a finite set S containing all ramified primes and all conductors involved. With the corresponding primes omitted upstairs,

$$L_K^S(s, \psi \circ N_{K/F}) = \prod_{\chi \in \widehat{G}} L_F^S(s, \psi\chi). \quad (2.1)$$

One can verify this directly on the absolute half-plane. If an unramified prime $\mathfrak{p}$ has Frobenius order f , it has $|G|/f$ primes above it, each of norm $(N\mathfrak{p})^f$. The values $\chi(\text{Frob}_{\mathfrak{p}})$ run through the f -th roots of unity, each $|G|/f$ times. Hence the two Euler factors agree. Meromorphic continuation gives (2.1) everywhere. The omitted Euler polynomials are nonzero in $\text{Re } s > 0$.

If all factors on the right are holomorphic at s_0 , a zero of any one of them gives a zero on the left. The only remaining possibility is a pole. By the standard Hecke pole classification [3], at most one factor has a pole at s_0 , and then $\psi\chi_0 = N_F^{it}$ and $s_0 = 1 + it$. Its pullback is N_K^{it} , so the left side and the χ_0 factor both have a simple pole. Taking meromorphic orders in (2.1),

$$-1 = -1 + \sum_{\chi \neq \chi_0} \text{ord}_{s_0} L_F^S(s, \psi\chi).$$

Every term in this sum is nonnegative, hence zero. No zero of another factor can cancel the pole. This proves the assertion on the whole half-plane, including its intersection with $\text{Re } s = 1$. $\square$

2.2 One family for each fixed infinity type

We now fix a totally imaginary number field K containing μ_{12} , and one unitary Hecke character ω of K . Let

$$\mathcal{F}_\omega = \{\theta \omega^e N_K^{it} : e \in \{-1, 0, 1\}, \theta \text{ finite order}, t \in \mathbb{R}\}. \quad (2.2)$$

All finite-order characters are included by $e = 0$. This family is closed under finite-order twisting, conjugation, and vertical translation of its L -functions. These are exactly the operations on live Hecke denominators in the argument. Extracted coefficients such as $\omega(\mathfrak{b})^3$ do not require a live background ω^3 ; Section 4 makes this distinction explicit.

The real parts of zeros are unchanged by the parameter t , so it may be omitted when taking suprema. After the starting bound is proved, set

$$B = B_{K,\omega} := \sup(\{1/2\} \cup \{\operatorname{Re} \rho : \eta \in \mathcal{F}_\omega, L_K(\rho, \eta) = 0, 1/2 \leq \operatorname{Re} \rho \leq 1\}). \quad (2.3)$$

The contradiction will use this one supremum for both the target and its companion denominators. It will never substitute a target's individual zero supremum for the common family boundary.

3 Transfer of the original constructions

All sums below use ideal norms. We write $q_{\mathfrak{a}} = N\mathfrak{a}$. The symbols u, n, p in the source-compatible formulas denote ideal indices with the specified finite class and unit sectors, or their allowed generators in the S -integer presentation of P , Section 3. They do not mean all algebraic integers of bounded absolute norm. The latter set can be infinite because of units.

Fix S large enough to include the archimedean places, the primes over 6, the fixed finite data, and representatives that trivialize the S -ideal class group. Use P 's character-equivariant integration and periodization modulo U_S^6 . In particular, the continuous unit phases of ω are retained. Only the finite power-class and ray tables are subjected to finite Fourier inversion. Zero extensions of residue symbols remain in every formula.

Proposition 3.1 (Quantitative field and type transfer). *Under Source Input 1.1, fix $K \supset \mu_{12}$ and a unitary Hecke character ω . The following three conclusions hold for the family (2.2). We use T , I , and P as names for these consequences.*

- T. The normalized cubic theta and Poisson constructions have the general-field form of P , Sections 3 and 5–7 and Appendix A, and the fully marked physical probe has the norm estimates of Q , Sections 5–16. This includes the actual unit-quotient profiles, finite local scalars, whole-index masks, all principal and zero orbits, and the full compensated Euler correction. More precisely, it supplies the normalized functions (6.8), the principal assertions (6.6)–(6.7), the low estimate (6.9), and the row decomposition (7.4) with the principal, small-row, and absolute large-row estimates specified in Section 7. The zero detector and witness construction of Q , Section 8, also hold with these norm scales. This transfer includes neither an improved row count nor a new zero-free boundary.*
- I. The canonical marked inverse estimate (4.3) holds with Q , Section 17's strict widths, allowed bounded coefficients, common masks, moving-puncture capacity, and finite-seminorm profile rules. Its row indexing admits sixth-power amplification with the original zero extensions. In particular, the independent scales in (4.5) are available; the amplification proof below explains this consequence of the unmarked estimate and ideal counting.*
- P. The paired ordinary fourth moment (5.3) holds for $3/4 \leq \kappa \leq 5/6$, with capacity $n_1 + n_2 + 6\kappa z \leq m + q$, the zero-slot endpoint, and the exceptional-row and common-mask conventions in Section 5. Its prime-cancellation premise is only that finite-order Hecke characters have the common boundary $(1 + \kappa)/2$. The corresponding reciprocal and prime bounds are used in the quantitative form of Q , Section 4.*

Each statement includes its finite-seminorm and polynomial-height assertions, with the real choices in the following paragraph.

Location of the proof. Proposition 3.2 establishes the general-field arithmetic calibration. Lemmas 3.3 and 3.4 give the exact fixed-type reflection, masked Poisson formula, and full additive Gram estimate. The finite marked reflection and the two canonical inductions are transferred in Appendix A. Lemma 4.1 verifies the permitted infinite-type backgrounds, and the proof of Proposition 5.1 checks ordinary reflection, prime pairing, and centered exceptional cancellation. These arguments use the full source reductions and supply every field-dependent replacement; they do not infer the moments from a large sieve alone. $\square$

3.1 The order of uniformity

On compact admissible ranges of real parameters, for every prescribed positive real loss, the width decrements, slot mesh, and finite slot system in Proposition 3.1 can be chosen depending on K , the fixed infinity type of ω , and those losses, before the later finite-order target twist and its conductor are selected. Adding finitely many excluded primes may change constants and thresholds but not these real choices. In the fourth moment the mesh is chosen before, and independently of, the fixed number of slots. Its eventual seminorm orders and constants may depend on that number.

Implied constants, finite seminorm and Sobolev orders, height degrees, height ceilings, and lower thresholds may depend on the full fixed target and S . The external tail orders are chosen after the height exponent. The transfer proofs give no additional degree-dependent or S -dependent power of the scale Z : fixed-degree unit logarithms consume a prescribed small loss, and higher-dimensional Poisson decay is obtained by increasing fixed derivative orders. The bounded induction depth depends on the strict real margins; finite sector multiplicities change constants and thresholds. Appendix A records the resulting order of choices.

Thus the argument does not require estimates uniform over all infinity types, all fields, or all conductors. It requires a common real saving within the later finite-order twists of one already fixed infinity type. These are the quantifiers used in Section 7.

3.2 General-field theta calibration

Proposition 3.2 (Field transfer of the theta construction). *Fix $K \supset \mu_{12}$. Retain the original cubic theta construction and local normalizations of [2, Proposition 5.1 and Appendix A]. Its arithmetic realization, Gaussian smoothing, annular reflection, and local Euler algebra apply over K to every fixed unitary target, after the conjugation choice of Lemma 3.3. These are consequences of that source construction, rather than additional general-field analytic assumptions.*

Proof. Choose S to contain the primes over 6, the trace-character ramification, the target's finite conductor, the source's fixed local data, and primes generating the ideal class group. Then $R = \mathcal{O}_{K,S}$ is principal. Adelic additive duality and the product formula identify R with its annihilator in K_S and give it covolume one in the self-dual normalization, exactly as in [2, Lemma 3.1]; the argument does not use a cyclotomic extension.

Put $[K : \mathbb{Q}] = 2r$ and $s_f = |S_f|$. For $\ell = 2, 3$, local Kummer theory and the S -unit theorem give

$$\dim_{\mathbb{F}_\ell} K_S^\times / K_S^{\times \ell} = 2s_f + 2r, \quad \dim_{\mathbb{F}_\ell} R^\times / (R^\times)^\ell = s_f + r.$$

The second space injects into the first: a unit locally an ℓ th power at S defines a Kummer extension unramified outside S and split at S , hence an everywhere unramified abelian extension classified by the trivial S -ideal class group. Hilbert reciprocity makes the image isotropic. Also $-1 = \zeta_{12}^6$ is a sixth power, so the sextic Hilbert pairing is alternating. The image is therefore Lagrangian, and it has an isotropic complement B . Combining the 2- and 3-parts gives the

allowed generators of [2, Lemma 3.2], unique modulo $\Gamma = (R^\times)^6$. Compactness of K_S^1/Γ supplies balanced representatives.

For the cubic Gauss correction set $\Lambda_K = \Lambda_{\mathbb{Q}(\sqrt{-3})} \circ N_{K/\mathbb{Q}(\sqrt{-3})}$. At a prime of residue degree f , Davenport–Hasse lifting gives $J_{q^f} = (-1)^{f-1} J_q^f$; the prime Jacobi identity consequently gives $\gamma_2(\mathfrak{p})^3 = -\Lambda_K(\mathfrak{p})$. Each complex coordinate of the relative norm contributes either z or $\bar{z}$, so the angular weights of Λ_K are ± 1 . The quadratic correction follows from the Weil-index product formula on the same isotropic complement. This proves the identities of [2, Lemma 3.4] over K , without a Galois or CM assumption. The Kazhdan–Patterson specialization used in [2, Appendix A] is over a general number field containing the required roots of unity: its $(2, 3, 2)$ scalar quotient and exceptional datum are local, while rational compatibility follows from reciprocity and the product formula. An everywhere-local cube is a global cube by the Kummer extension and Chebotarev argument given there. Thus no cyclotomic restriction enters this specialization of the retained construction.

For an infinite-order target, only its smooth local unit restrictions are cut off at finite places. They have open kernels even if their uniformizer values have infinite order. Lemma 5.3 of [2] therefore supplies a common cutoff inside these kernels and the sixth powers. The target itself is never expanded in a finite Fourier table. At infinity use Lemma 3.3, keeping all relative radial parameters. Its exact orbit substitution puts the whole target value on the completed ideal index, and its Mellin calculation supplies the Gaussian and arbitrary annular tests. In particular no extra norm phase is introduced in the physical scale monomial. The exterior residue fields have $q_{\mathfrak{p}} \equiv 1 \pmod{12}$, so all remaining local digit, Gauss, and Euler computations of [2, Sections 6–7] are unchanged. $\square$

3.3 The archimedean interface over a general field

Let $K \supset \mu_{12}$ be fixed, with $[K : \mathbb{Q}] = 2r$. Use the theta representation, rational splitting, and coefficient normalizations constructed in the arithmetic part of Proposition 3.2. The following local and norm-sheet calculation uses [2, Proposition 5.1 and Appendix A], retaining every fixed angular and relative radial parameter.

Put $R = \mathcal{O}_{K,S}$ and $\Gamma = (R^\times)^6$, with S enlarged as in [2, Section 3]. Let $\mathcal{L}$ be the actual unitary S -period character, including the Gauss correction, so $\mathcal{L}(\Gamma) = 1$. At a complex place write

$$\mathcal{L}_v(z) = \omega_{m_v}(z) |z|_v^{i\tau_v}, \quad \omega_m(z) = (z/|z|_{\text{ord}})^m, \quad m_v \in \mathbb{Z}, \quad \tau_v \in \mathbb{R}.$$

The relative parameters τ_v are retained; a single global norm twist need not remove them. The Gauss correction has angular exponents $b_v \in \{1, -1\}$ by [2, Lemma 3.4]. In the ideal-character convention, the period weights are $m_v = k_v - b_v$ for target angular weights k_v . Replacing the target by its conjugate if necessary ensures $m_v \neq 0$ at at least one place: the two vectors $(k_v - b_v)_v$ and $(-k_v - b_v)_v$ cannot both vanish. Finite-order children do not change these weights.

Lemma 3.3 (Fixed-type norm-sheet reflection). *Assume the preceding nonzero angular condition. The theta period has an exact reflected transform for every smooth test supported in a fixed compact subinterval of $(0, \infty)$. Its archimedean transform, including each fixed finite- S branch, satisfies*

$$|(x\partial_x)^h \Phi_{\kappa,V}(x)| \ll_{K,\mathcal{L},A,h} \mathcal{S}_J(V) \min\{x^{1/4}, x^{-A}\}, \quad J > 4rA + h + 1,$$

where $\mathcal{S}_J$ is a finite smooth seminorm on that interval. For the fixed Gaussian test the same Mellin identity applies; its constants may depend on the fixed type. A moving norm phase $V(x)x^{iT}$ costs at most $(1 + |T|)^J$ in this estimate.

Proof. At a complex place let ϕ° be spherical and choose the additive coordinate as in [2, Section 5.2]. The holomorphic and antiholomorphic upper-root operators $E, \bar{E}$ can be normalized with the same scalar so that they multiply the Whittaker function by $z, \bar{z}$, respectively; this is (5.10)

of [2]. Take $\phi_m = E^m \phi^\circ$ for $m \geq 0$, and $\phi_m = \bar{E}^{-m} \phi^\circ$ for $m < 0$. With $a_v = (|m_v| + 1)/2$, the local Whittaker function is

$$W_{\phi_{m_v}}(a(z)) = |z|_v^{a_v} \omega_{m_v}(z) K_{1/3}(c_v |z|_v^{1/2}), \quad c_v > 0,$$

after a common spherical normalization. The complexified compact Lie algebra annihilates ϕ° ; in particular $(F - \bar{E})\phi^\circ = 0$. Since F and $\bar{E}$ commute, $F^j \phi^\circ = \bar{E}^j \phi^\circ$. Write $\mathcal{K}_v$ for the maximal compact subgroup. Then $w \in \mathcal{K}_v$ and $\text{Ad}(w)E = -F$, giving the exact identity

$$\pi(w)\phi_m = (-1)^{|m|}\phi_{-m}.$$

Thus each complex place is treated independently; no common angular weight or single global derivative is used.

At the finite places choose the Whittaker cutoff $\mathbf{1}_U$ from [2, Lemma 5.3], with U contained in the sixth powers and in $\ker \mathcal{L}$. Write $X_u = \{y \in K_S^\times : |y|_S = u\}$ with the sheet measure of [2, (5.15)]. The unreflected profile is

$$V_0(u) = \int_{X_u} \mathbf{1}_U(y_f) \prod_{v|\infty} t_v^{a_v - i\tau_v} K_{1/3}(c_v \sqrt{t_v}) d_u^\times y, \quad t_v = |y_v|_v.$$

Using $\hat{f}(s) = \int_0^\infty f(x)x^s dx/x$, absolute Fubini and the Bessel integral give, initially for $\Re s > -1/3$,

$$\hat{V}_0(s) = C_0 A_0^s \prod_{v|\infty} \prod_{\epsilon=\pm 1} \Gamma(a_v + s - i\tau_v + \epsilon/6), \quad C_0 \neq 0, \quad A_0 = \prod_v 4/c_v^2 > 0.$$

For a reflected branch κ , the character in the profile is $\mathcal{L}$, rather than $\mathcal{L}^{-1}$, and the vector is $\pi(w)\phi$. Consequently the exact Mellin multiplier is

$$\mathcal{R}_\kappa(s) := \frac{\hat{H}_\kappa(s)}{\hat{V}_0(-s)} = C_\kappa A_0^{2s} B_\kappa(s) \prod_{v|\infty} \prod_{\epsilon=\pm 1} \frac{\Gamma(a_v + s + i\tau_v + \epsilon/6)}{\Gamma(a_v - s - i\tau_v + \epsilon/6)}. \quad (3.1)$$

Here B_κ is a finite sum of finite-place Mellin factors. For every $\delta < 1/3$, (5.13) of [2] makes it holomorphic on $\Re s > -\delta$ and bounded on each closed vertical substrip. The original finite factor is the nonzero constant $\text{vol}(U)$; no arbitrary finite Mellin factor is divided out. Gamma poles have real part at most $-a_v + 1/6 \leq -1/3$. Taking $1/4 < \delta < 1/3$, the multiplier is therefore holomorphic through the line $\Re s = -1/4$. Stirling's formula cancels the exponential factors separately at each place, since the imaginary arguments in each quotient are opposite. On a fixed vertical line in this region,

$$|\mathcal{R}_\kappa(\sigma + it)| \ll_{K, \mathcal{L}, \sigma} (1 + |t|)^{4r\sigma}.$$

For completeness, this multiplier belongs to an exact period identity, not merely a formal local transform. One integrates on X_u/Γ , which is compact by the S -unit theorem. Rotating at a place with $m_v \neq 0$ kills both constant terms, because their inducing character is radial; this is the argument of [2, (6.8)]. The absolute bounds of [2, Lemma 5.4] apply unchanged to the fixed unitary character. Choose a measurable fundamental domain $\Omega_u \subset X_u$ for X_u/Γ . Orbit unfolding is the identity

$$\sum_{\gamma \in \Gamma} \int_{\Omega_u} \mathbf{1}_B([y]) \mathcal{L}(y)^{-1} f(\gamma A y) d_u^\times y = \mathcal{L}(A) \int_{X_{q_A u}} \mathbf{1}_B([A]^{-1}[v]) \mathcal{L}(v)^{-1} f(v) d^\times v,$$

as in [2, (6.13)]. On the unreflected cutoff $[v] = 1$ and $[A] \in B$, so the remaining class indicator is one. Thus ideals are counted once, and no count of their infinitely many unit associates is

substituted for a norm-sheet integral. The two expansions give the completed Mellin functional equation. Its continuation is justified by the rapid decay at opposite ends after constant-term removal; reciprocal gamma factors are entire. The convergent expansions on opposite half-planes and Phragmen–Lindelof then give polynomial growth on fixed strips.

Mellin inversion now yields the reflected kernel

$$\Phi_{\kappa,V}(x) = \frac{1}{2\pi i} \int_{(\sigma)} \widehat{V}(-s) \mathcal{R}_{\kappa}(s) x^{-s} ds.$$

Shift to $\sigma = -1/4$ or $\sigma = A$ and integrate by parts J times in the Mellin transform of V . This proves the estimate. Finally $\mathcal{S}_J(Vx^{iT}) \ll (1 + |T|)^J \mathcal{S}_J(V)$. One keeps this fixed-type reflection and changes the test; one does not rebuild Gaussian deconvolution kernels for each T . $\square$

The field degree increases the fixed seminorm order J , but adds no norm-length exponent to this lemma. Once all internal seminorm orders have been fixed, a height $T_1 = Z^\tau$ with sufficiently small fixed $\tau > 0$ absorbs its polynomial cost; the external tail order is chosen afterwards. The Poisson and row estimates are transferred next, and the canonical moment reductions are completed in Appendix A. These use the same finite-seminorm profile class as the reflection lemma.

3.4 Poisson and additive correlation

Lemma 3.4 (Norm-form Poisson and the additive Gram estimate). *Fix a number field $K \supset \mu_{12}$ and fixed arithmetic data as in P , Section 3. Let $R = \mathcal{O}_{K,S}$, and choose a finite-index power subgroup $\Gamma \subset R^{\times 6}$ on which the relevant finite characters are trivial. Put $T = K_S^1$. The masked Poisson operations in Q may be performed on Γ -orbits with ideal norms. Their scale factors and the additive Gram bound Q , (15.4), are unchanged. More precisely, for the additive polynomial of Q , Section 6, with its specified finite ray coefficients and a fixed compact additive row shape, put $P_a = Y^2/Q \geq 1$. Then*

$$\sum_m |A_m(Y)|^2 \ll_{K,S,\epsilon} (1 + |\nu|)^J \frac{Q}{Y} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y} \right) Z^\epsilon.$$

Here $\nu \in \mathbb{R}$ is the norm-power height in the additive test, the row shape has norm scale Q , the scales lie in fixed polynomial ranges, and J is a finite fixed seminorm order. No arbitrary row-dependent arithmetic coefficient is allowed. The real loss ϵ is prescribed before the fixed target and its finite conductor; constants, J , and thresholds may depend on those fixed data.

Proof. The quotient T/Γ is compact. Choose a smooth, compactly supported unit partition on T , with the finite-place components locally constant, whose translates by Γ sum to one. This represents orbit sums by lattice sums without counting infinitely many associates. If ℓ_x is the positive norm section, define

$$W_\phi(x) = \int_T \phi(\ell_x v) d^\times v, \quad \widetilde{W}_\phi(x) = \int_T \widehat{\phi}(\ell_x v) d^\times v.$$

Every compact smooth annular norm test has such a lift, by multiplying it by a fixed compact shape function of integral one. For a primitive exterior residue character χ of conductor $\mathfrak{m} = mR$, the nonprincipal masked formula is

$$\sum_{[k] \neq 0} \chi(k) \mathbf{1}_{(k,\mathfrak{R})=1} W_\phi(q_k/L) = \frac{L\gamma(\chi)}{\sqrt{N\mathfrak{m}}} \sum_{\mathfrak{d} \mid \text{rad } \mathfrak{R}} \frac{\mu(\mathfrak{d})\chi(d)}{N\mathfrak{d}} \sum_{[h] \neq 0} \bar{\chi}(h) \widetilde{W}_\phi\left(\frac{Lq_h}{N\mathfrak{d}N\mathfrak{m}}\right).$$

Indeed, expand the mask, substitute $k = dk'$, and apply additive Poisson on the self-dual R -lattice while integrating on the compact quotient T/Γ . The finite transform is the ordinary primitive

Gauss transform. Haar translation removes the norm-one part of dm , so the transformed kernel is common and has no moving modulus shape. Fixed local sectors are separated before this operation. For a general periodic row coefficient one retains its full finite Fourier transform instead of replacing it by a primitive character. The dual zero is kept separately; for the primitive principal character it is

$$L \operatorname{vol}(T/\Gamma) \hat{\phi}(0) \sum_{\mathfrak{d} \mid \operatorname{rad} \mathfrak{R}} \frac{\mu(\mathfrak{d})}{N\mathfrak{d}}.$$

Any included primal zero is the separate compact-quotient scalar $\operatorname{vol}(T/\Gamma) \phi(0)$ with its actual mask. It is never interpreted as $W_\phi(0)$.

The general periodic version, including both zero orbits. Let $f : R/\mathfrak{c} \rightarrow \mathbb{C}$ be a periodic function such that $f(\gamma a) = f(a)$ for every $\gamma \in \Gamma$; every fixed local sector and every original zero extension is retained in f or in the fixed local profile. For $\mathfrak{d} \mid \operatorname{rad} \mathfrak{R}$ define the unnormalized finite transform

$$\hat{f}_d(h) = \sum_{a \bmod \mathfrak{c}} f(da) e(ha/c), \quad \mathfrak{c} = cR.$$

Use the additive-sign convention consistent with the chosen lattice Fourier transform. Here $[k], [h] \in \Gamma \backslash (R \setminus \{0\})$, $q_k = N(kR) = |k|_S$, and c, d are chosen exterior generators. The exact orbit formula is

$$\sum_{[k] \neq 0} f(k) \mathbf{1}_{(k, \mathfrak{R})=1} W_\phi(q_k/L) = \frac{L}{q_c} \sum_{\mathfrak{d} \mid \operatorname{rad} \mathfrak{R}} \frac{\mu(\mathfrak{d})}{q_d} \sum_{[h] \neq 0} \hat{f}_d(h) \widetilde{W}_\phi\left(\frac{Lq_h}{q_d q_c}\right) + \mathcal{Z}_f,$$

where

$$\mathcal{Z}_f = \frac{L \operatorname{vol}(T/\Gamma) \hat{\phi}(0)}{q_c} \sum_{\mathfrak{d} \mid \operatorname{rad} \mathfrak{R}} \frac{\mu(\mathfrak{d})}{q_d} \hat{f}_d(0) - \operatorname{vol}(T/\Gamma) \phi(0) f(0) \sum_{\mathfrak{d} \mid \operatorname{rad} \mathfrak{R}} \mu(\mathfrak{d}).$$

The last term is absent for an annular lift. The mask is expanded before the finite transform: for general f one transforms $f(da)$, not a fictitious multiplicative factor $f(d)f(a)$. Since every $\gamma \in \Gamma$ is a unit modulo the exterior modulus, a residue substitution shows that $\hat{f}_d$ is also Γ -invariant. Both nonzero sums are therefore genuine orbit sums. Specializing to a primitive character gives $\hat{f}_d(h) = \chi(d) \sqrt{q_c} \gamma(\chi) \bar{\chi}(h)$ and recovers the preceding character formula. For the primitive principal character $\mathfrak{c} = R$, the dual-zero formula specializes to the displayed principal scalar.

On positive norm sheets, product Schwartz bounds give arbitrary power decay at infinity and at most $(1 + |\log x|)^b$ at zero, where b depends only on the fixed field and finite data. Euler derivatives have the same logarithmic-degree bound. Thus the additional logarithms consume a prescribed small real loss, with only the threshold depending on the data. A positive Gaussian/compact-open lift majorizes the entire row ball.

For the Gram bound follow the full-correlation proof of Q, Proposition 15.2, retaining its collision zeros. Write $s_i = Cn_i$, $(n_1, n_2) = 1$. The first Poisson transform has precisely the prefactor Q/Y^3 : the lattice denominator $q_{s_1} q_{s_2}$ cancels the same factor in the complete Gauss correlation. The zero frequency forces $n_1 = n_2 = 1$ and contributes $O(Q/Y)$. Insert the complete divisor identity for coprimality before extending the coefficient, and write $n_i = dm_i$. Its bounded extension is used with all original shared-prime zeros. After reciprocity the exterior period is

$$\mathfrak{r} = \operatorname{lcm}(\operatorname{rad} C, \operatorname{rad}(k)), \quad q_{\mathfrak{r}} \ll_{\text{fixed}} q_C q_k,$$

with the fixed local data included in the fixed profile or sectors. For a nonexceptional k , namely one with a valuation not divisible by six at a prime outside C and the fixed support, the normalized two-variable finite transform has size at most q_C and its zero coefficient vanishes. These are the same finite-field and CRT identities as in Q; the residue-field cardinality simply becomes $q_{\mathfrak{p}}$.

Keep one joint two-column profile, rather than separately chosen row-dependent Fourier densities. On a shell $q_C q_k / P_a \asymp \mathcal{R}$, this profile has homogeneous seminorms $O((1 + |\nu|)^J \mathcal{R}^{-B})$. Split $q_C \asymp C_0$ and $q_d \asymp D_0$ into dyads. With $N = Y/(C_0 D_0)$, use the unit partition in both columns and balance an allowed generator of $\mathfrak{r}$ by P, Lemma 3.2. Its normalized shape stays in a fixed compact set. Two-column lattice Poisson gives

$$\left| \sum_{m \in R^2} a_{C,d,k}(m) W_{\mathcal{R}}(m/\ell_N) \right| \ll (1 + |\nu|)^J \mathcal{R}^{-B} q_C N^2 \min\{1, (q_{\mathfrak{r}}/N)^A\}.$$

For $N \geq q_{\mathfrak{r}}$ the zero frequency is absent. A nonzero dual vector has a large archimedean coordinate; taking the fixed Schwartz order sufficiently large in terms of $A[K : \mathbb{Q}]$ gives the displayed decay. For $N < q_{\mathfrak{r}}$ use the compact profile point count; a nonempty subunit annulus is bounded below by a fixed constant. Thus field degree changes the derivative order, not the exponent of N .

Ideal counting bounds the nonexceptional frequencies on this shell by $O(\mathcal{R} P_a / C_0)$, up to a fixed-degree logarithm for representatives in compact additive shapes. The same positive dyadic sum as in Q is

$$\sum_{i,j \geq 0} 2^{-i-j} \min\{1, (2^{i+j}/\Lambda)^A\} \ll_A \frac{1 + \log(2 + \Lambda)}{\Lambda}, \quad \Lambda = Y/(\mathcal{R} P_a).$$

After the shell sum this gives $(Q/Y)(P_a^2/Y)Z^{\epsilon}$. Exceptional nonzero frequency ideals have the form $\mathfrak{a}^6 \mathfrak{e}$, with $\mathfrak{e}$ supported on C and the fixed support. There are q_C^{ϵ} possibilities for $\mathfrak{e}$ and $O((\mathcal{R} P_a / C_0)^{1/6})$ possibilities for $\mathfrak{a}$. Representatives of one exterior ideal in the compact additive shapes are S -units in a logarithmic box, hence contribute only a fixed-degree logarithm. The corresponding term is $(Q/Y)P_a^{1/6} Z^{\epsilon}$. Adding the diagonal gives the claim. $\square$

Application to a fixed infinite type. In the physical probe the target character acts on the completed theta column cn^3 , not on the additive row variable m or the additive polynomial A_m . Consequently the preceding Gram argument is literally unchanged for a genuine unitary target. Its continuous infinity phase is handled by the fixed-type theta reflection lemma; compensation contributes only the bounded column/slot coefficient $\overline{\psi(p)}$. The completed-index masks, finite correlations, local Euler algebra and subset scale coefficients remain the same. The arbitrary-annular reflection has the same two-ended kernel bound, with higher fixed seminorm and height orders. Therefore the homogeneous common-profile separation and the positive row-energy calculation of Q, Sections 14–15, retain their norm exponents. At the original geometry the low estimate is $Z^{3/16+\epsilon}$. The high Euler factors use only unit-modulus local target values; their prime sums converge in the same regions over a fixed number field. The principal residue constant is nonzero and may be calibrated by one fixed scalar, as in P, Section 8. None of these steps imposes a new zero-free hypothesis.

4 The canonical family and the starting bound

Fix a finite group Θ of finite-order ray characters containing the fixed finite Gauss and reciprocity corrections. Enlarge the fixed set S to contain every fixed conductor and excluded prime. For a fixed target $\psi \in \mathcal{F}_{\omega}$, its genuine infinite-order part is not made a member of Θ . Global norm twists are placed in the tests, and the finite-order cases use the original finite class. For a finite-order target, enlarge Θ to contain ψ and absorb ψ into the finite presentation; in the paired moment this case may always be written with $\omega_0 = 1$. Writing $\chi_u(n) = \chi_n(u)$ for the physical sextic presentation, the actual row family is

$$\mathcal{X}_u = \{\theta \chi_u, \theta \bar{\chi}_u : \theta \in \Theta\} \cup \{\psi \theta \bar{\chi}_u, \bar{\psi} \theta \chi_u : \theta \in \Theta\}. \quad (4.1)$$

It is closed under conjugation. For each physical row it contains finitely many characters, although some have infinite order.

4.1 Preservation of the fixed infinity type

For a presentation $\nu\chi_u^{\epsilon_\chi}$, the initialization in Q, (17.68), conjugates the whole polynomial when $\epsilon_\chi = +1$. Its canonical background is thus

$$\nu_{\text{can}} = \nu^{-\epsilon_\chi}. \quad (4.2)$$

The two conjugate genuine presentations in (4.1) both give a background in $\psi\Theta$; finite-order presentations give one in Θ .

Lemma 4.1 (Directed canonical reduction). *The canonical reductions of Q, Section 17, preserve the class*

$$\nu_{\text{can}} \in \Theta \cup \psi\Theta.$$

No live background with a new power of the genuine character is introduced. Together with the character-equivariant theta reflection and the canonical transfer in Appendix A, this is the entire infinite-type family needed by the marked inverse estimate.

Proof. We check the places where finite order could matter. In the first Poisson step, Q, (17.22)–(17.30), write $n_i = A_i C u_i$. On the original support,

$$\nu(n_1)\overline{\nu(n_2)} = \nu(A_1)\overline{\nu(A_2)}\nu(u_1)\overline{\nu(u_2)}.$$

The common factor cancels because $|\nu(C)| = 1$. Keep the last two factors outside the finite Gauss-ray table. Expanding that table then gives backgrounds $\nu\vartheta_i$, $\vartheta_i \in \Theta$; no finite Fourier expansion of ν is being used. In the second Poisson step, Q, (17.40)–(17.55), the common extracted factor again contributes $|\nu\vartheta_i(v')|^2 = 1$. A child background is $\nu\vartheta_i\vartheta$, still in the same class.

Cube extraction in Q, (14.2) and (17.10), gives $\nu(b)^3$ on the cube variable. This is an allowed bounded coefficient on that variable, not a replacement of the live residual character by ν^3 . Divisor extraction and the priority identities for marks keep their original masks and product coefficients.

The remaining use of the background is the completed theta terminal in Q, (14.14). Over K this is supplied by Lemma 3.3 and the finite marked reflection in Appendix A. The finite table is kept separate from the fixed archimedean phase. Its decay and finite-seminorm bounds are exactly those used in the transferred canonical proof. Thus the finite tables, masks, and live background are all preserved, including at the reflected terminal. $\square$

The resulting inverse estimate I has the following precise form. With the source-compatible prime slots $Q_{u,i}$, total slot length z , and a fixed polynomial-length inverse sum

$$\mathcal{M}_u(D; W) = D^{-1/2} \sum_n \mu(n) \nu(n) \chi_n(u)^{\epsilon_\chi} W(q_n/D),$$

take only presentations from (4.1), equivalently $\nu^{-\epsilon_\chi} \in \Theta \cup \psi\Theta$. The resulting estimate is

$$\sum_{0 < q_u \ll Z^m} \left| \mathcal{M}_u(Z^r; W) \prod_i Q_{u,i} \right|^2 \ll Z^{m+\varepsilon}, \quad r + 2z < m, \quad 2r + 8z < 3m. \quad (4.3)$$

Each strict inequality has a fixed positive margin. Slot coefficients are bounded and fixed independently of the row; supports are disjoint before masks are imposed. All products retain the same coprimality mask. Only the rowwise profile choices permitted by Q, Lemma 4.5, are allowed. At initialization the masks come only from the fixed set S and the displayed presentation zeros. Moving punctures generated inside the canonical reduction must satisfy Q, (17.2), with their capacity counted there. An additional freely chosen moving mask is not allowed, even if held fixed during one row sum.

4.2 A single-row amplification

Proposition 4.2. *Under Source Input 1.1, the transferred inverse estimate implies, for every $\eta \in \mathcal{F}_\omega$,*

$$L_K(s, \eta) \neq 0 \quad (\operatorname{Re} s > 11/12). \quad (4.4)$$

This implication uses no prior zero-free bound for genuine infinite-order characters.

Proof. Remove a global norm twist and use the permitted orientation of the fixed infinite type; finite-order cases are included. Use $\nu = \eta$ and $\epsilon_\chi = -1$ in the inverse polynomials. The first two-scale statement of Q, Lemma 17.6, applies to independent $U, D \geq 1$ and gives

$$\sum_{u \sim U}^{\text{sixth-power-free}} |\mathcal{M}_u(D; W)|^2 \ll \max\{U, U^{1/6} D^{5(1+c)/6}\} (UD)^\epsilon \quad (c > 0). \quad (4.5)$$

Take a fixed bounded row shell containing $u = 1$. This is the independent-scale assertion, not the later specialization in which $D = U^r$ and r is bounded.

For completeness, the amplification in this special case is short. Put $H = D^{1+c}$ and $P = H^{1/6}$. The no-slot case of (4.3), with base H , bounds the row energy up to H by $H(HD)^\epsilon$, also after the scale supremum in Q, Lemma 4.5. For each allowed ideal a with $q_a \leq P$, the exact identity is

$$\mathcal{M}_1(D; W) = \sum_{d|\operatorname{rad}(a)} \mu(d) \eta(d) q_d^{-1/2} \mathcal{M}_{a^6}(D/q_d; W). \quad (4.6)$$

It follows from $\chi_n(a^6) = \mathbf{1}_{(n,a)=1}$. There are $\gg_{K,S} P$ integral ideals a in an allowed fixed class with $q_a \leq P$. Their sixth powers are distinct ideal rows of norm at most H , and the divisor coefficient mass is a small power. Weighted Cauchy and averaging give

$$|\mathcal{M}_1(D; W)|^2 \ll H/P \cdot D^\epsilon = D^{5(1+c)/6+\epsilon}.$$

Letting c and the preliminary loss be sufficiently small proves

$$\sum_{(n,S)=1} \mu(n) \eta(n) W(q_n/D) \ll_{\eta, W, \epsilon} D^{11/12+\epsilon}. \quad (4.7)$$

Choose a smooth dyadic partition $\sum_{j \geq 0} W(q/2^j) = 1$ for $q \geq 1$. On a compact subset of $\operatorname{Re} s > 11/12$, the tests $W_s(y) = W(y)y^{-s}$ have bounded relevant seminorms. Thus

$$G_\eta(s) = \sum_{j \geq 0} 2^{-js} \sum_{(n,S)=1} \mu(n) \eta(n) W_s(q_n/2^j)$$

converges locally uniformly there by (4.7). For $\operatorname{Re} s > 1$ it equals $1/L_K^S(s, \eta)$. The meromorphic identity $L_K^S(s, \eta) G_\eta(s) = 1$ proves the claim. Deleted Euler factors have no zeros in $\operatorname{Re} s > 0$.

For two varying scales the same proof averages over (u, a) : valuation modulo six recovers the sixth-power-free ideal u from ua^6 , and the quotient recovers a . The finitely many class and unit sectors cost only a fixed factor. This proves (4.5) from the same transferred row-energy estimate.

The starting argument uses only T and I, corresponding to Q, Section 17 and its theta terminals. It uses neither the fourth moment of Section 18 nor prime cancellation from a genuine-character zero-free region. $\square$

5 The paired fourth moment

Proposition 4.2 gives $B \leq 11/12$ in (2.3). In the contradiction $B > 7/8$, put

$$\Delta = B - \frac{7}{8}, \quad \kappa = 2B - 1 = \frac{3}{4} + 2\Delta, \quad 0 < \Delta \leq \frac{1}{24}. \quad (5.1)$$

Every finite-order Hecke character lies in $\mathcal{F}_\omega$. Consequently the prime premise of the transferred fourth moment is already supplied by the definition of B . A finite-order 7/8 theorem is not being assumed, and κ is not replaced by 3/4 in the actual fourth-moment capacity.

Let ξ_u be the finite moving presentation in $\mathbf{Q}$, Section 18, including its fixed finite-ray corrections and all zero extensions. Let ω_0 be either 1, ψ , or $\bar{\psi}$, and put $\lambda_u = \omega_0 \xi_u$. For a positive-length prime slot, take the paired coefficient $\omega_0^{-1} \vartheta_i$, $\vartheta_i \in \Theta$. Its actual prime polynomial is

$$Q_{u,i} = P_i^{-1/2} \sum_p \xi_u(p) \vartheta_i(p) W_i(q_p/P_i). \quad (5.2)$$

Consequently the character that needs cancellation at primes is finite order.

Proposition 5.1 (Paired fourth moment). *Under Source Input 1.1, retain the row, common-mask, and fixed moving-support hypotheses of $\mathbf{Q}$, Lemma 18.1, and write its effective width as $M = m + q$. For a sufficiently fine fixed slot mesh,*

$$\sum_{u \in \mathcal{R}_z} |S_{\lambda_u}(n_1; W_1) S_{\lambda_u}(n_2; W_2) \prod_i Q_{u,i}|^2 \ll Z^{M+\varepsilon}. \quad (5.3)$$

Here S denotes the source's centrally normalized plain sum. For positive total slot length z , it suffices that

$$n_1 + n_2 + 6\kappa z \leq M,$$

and $\mathcal{R}_z$ excludes rows whose finite inducing character ξ_u^* belongs to Θ . For $z = 0$, the bounded nonnegative lengths n_1, n_2 are unrestricted, and only a principal actual inducing character λ_u^* is excluded. The constants have the source's finite-seminorm and fixed polynomial-height dependence.

Compatibility checks. We record the changes in the source's fourth-moment reduction. The canonical recursion and its field-dependent counting steps are transferred in Appendix A. The following checks supply its fixed-type reflection, paired prime premise, and centered exceptional terminal.

First, ordinary Hecke reflection is available for λ_u . Writing its fixed complex local parameters as (k_v, τ_v) , the gamma quotient is, up to fixed exponential scales,

$$\prod_{v|\infty} \frac{\Gamma(s + |k_v|/2 + i\tau_v)}{\Gamma(1 - s + |k_v|/2 - i\tau_v)}.$$

Its poles have nonpositive real part and it has polynomial growth on fixed vertical strips away from its poles. The finite conductor factor is $Q^{1/2-s}$, with $Q = |d_K| \text{Nf}_{\lambda_u}$, so the dual norm length is still Q/X [3]. Coincident poles can produce powers of $\log x$ at the small end of the reflected profile. The central annular coefficients are therefore bounded by $2^{-j/2}(1+j)^A$, still summable. The common-mask identities $\mathbf{Q}$, (18.4)–(18.8), must be applied before a primitive inducing character is substituted.

Second, both finite Fourier transformations in $\mathbf{Q}$, Sections 18.4–18.6, act on the finite row character. Keep the factors $\omega_0(n_1) \overline{\omega_0(n_2)}$ on the two columns outside each finite table. Extraction multiplies them by unit-modulus scalars; it leaves a single ω_0 on each surviving plain character. Its matching inverse remains on the slot coefficient. Conjugating a reflected expression is handled in two ways. After reflecting a single plain factor, conjugate that factor and its test to recover the original character inside its absolute square. A change of the whole row orientation conjugates both plain factors and every slot, including their coefficients and tests. Thus every live prime polynomial continues to have a finite actual character as in (5.2). $\mathbf{Q}$'s prime estimate uses the common boundary B and hence $\kappa = 2B - 1$. The transferred norm-form capacity is $n_1 + n_2 + 6\kappa z \leq M$. The dependence on Δ must be retained in the later row comparison.

Third, the exceptional finite inducing family is handled on unmasked functions before inserting a moving radical. For each fixed $\vartheta \in \Theta$, Hecke continuation and a Mellin shift give, uniformly for $X > 0$,

$$\sum_n (\omega_0 \vartheta)(n) W(q_n/X) (q_n/X)^{it} = c_{\omega_0 \vartheta} X I_W(t) + O(p_J(W)(1+|t|)^A).$$

After global norm twists have been removed, the coefficient $c_{\omega_0 \vartheta}$ is zero unless the character is principal; $I_W(t) = \int_0^\infty W(y) y^{it} dy$. For $X \geq 1$ the error follows by shifting the unmasked L -function to $\operatorname{Re} s = -1$ and is in fact $O(X^{-1})$ with the same type of seminorm bound. For $X < 1$ bounded ideal counting suffices.

Insert the common radical R only afterwards by inclusion–exclusion. The main coefficient is multiplied by $\sum_{d|R} \mu(d) (\omega_0 \vartheta)(d)/q_d$, independently of the scale; the error grows by at most Z^ε when q_R is polynomially bounded. Keep the common kernel and mask through the two equal-product rectangles before separating the centered expression. Their principal main terms cancel exactly as in Q, Section 18.7. For a genuine infinite-type ω_0 the residue is zero, giving a stronger bound. Thus this exceptional calculation adds no degree-dependent power of the moving norm. It preserves the exceptional-row rule stated above. More quantitatively, if the four scales in the two rectangles are at least Z^ϱ , $\varrho \geq 0$, and have common product T , the unnormalized error after cancellation is $O(Z^\varepsilon T Z^{-\varrho} (1+|t|)^A)$. Central normalization divides by $T^{1/2}$, giving $O(Z^\varepsilon T^{1/2} Z^{-\varrho} (1+|t|)^A)$, the saving used in Q, (18.47). $\square$

6 Compensation and the principal factor

Let $I_{\psi;D}$ denote the physical probe in Q with target ψ and the mark $\mathbf{1}_{D|cn^3}$ on its whole completed index. Proposition 3.2 supplies its character-equivariant norm-form realization over K . At a good prime p , $Q = q_p$, the compensation is

$$\overline{\psi(p)} I_{\psi;p}(X, Y, ZQ) - Q^{-3/2} I_{\psi}(X/Q, Y/Q, Z). \quad (6.1)$$

For disjoint prime windows $\mathcal{P}_i$ and $p_J = \prod_{i \in J} p_i$, its full form is

$$\begin{aligned} I_{\psi, \text{mod}}(Z) &= \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, J_0\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\psi(p_{J^c})} \\ &\quad \cdot I_{\psi; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \end{aligned} \quad (6.2)$$

The target still acts on the whole completed index. All extra target factors have absolute value one, so the subset rescaling coefficients have exactly the sizes used by the low estimate in T.

The physical Mellin monomial is $X^{1/2-z} Z^{s+z-1} Y^{w-1}$. Marking at p and replacing Z by ZQ contributes $Q^{s+z-1} P_p^*$; the term with coefficient $Q^{-3/2}$ and scales $(X/Q, Y/Q)$ contributes $Q^{z-w-1} P_p$. After extracting Q^{z-1} , their difference is $\bar{\psi}(p) Q^s P_p^* - Q^{-w} P_p$. This verifies the scale and phase in (6.1).

For the remaining local calculation put

$$V = Q^{-6z}, \quad W = \chi_u(p) Q^{-w}, \quad D = \psi(p) \bar{\chi}_u(p) Q^{-s}.$$

If P_p is the local sum and P_p^* its marked part, then

$$P_p - P_p^* = (1 - V)^{-1} + W(1 - W)^{-1}, \quad B_p = \overline{\psi(p)} Q^s P_p^* / P_p - Q^{-w}.$$

Since $\overline{\psi(p)} Q^s D = \bar{\chi}_u(p)$, the main prime factor is

$$\mathcal{Q}_i(u; z) = - \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{z-1} \bar{\chi}_u(p). \quad (6.3)$$

This is exactly the finite actual prime character needed in Proposition 5.1. No condition $\psi(p) = 1$ is imposed on the prime windows.

The moment pairing can be made explicit. Orient a genuine detector witness as $\lambda_u = \psi\theta\bar{\chi}_u$. For the inverse moment, use direction -1 and relative slot coefficient $(\psi\theta)^{-1}$, which is bounded and independent of the row. For the plain fourth moment, conjugate both plain witnesses and all selected prime slots together. The resulting plain character is $\bar{\psi}\bar{\theta}\chi_u$, and its relative slot coefficient is $\psi\theta$. Thus in Proposition 5.1 one has $\omega_0 = \bar{\psi}$, $\xi_u = \bar{\theta}\chi_u$, and the required coefficient $\omega_0^{-1}\theta$. The actual slot is χ_u with the conjugated profile. The other presentations in (4.1) reduce to these by conjugation and finite twists. Test heights only change sign, so the source profile estimates still apply.

Contour shifts use the full local correction, without dividing by a possibly vanishing P_p . For example the replacement factor can be written

$$G_p = \frac{(\overline{\psi(p)}Q^s - Q^{-w})(1-V)(1-W)P_p^* - Q^{-w}(1-VW)}{1-D}.$$

The two Euler regions and the joint error-slot estimates in Q, Section 16, use only unit-modulus local target values. Their local algebra is unchanged over K . The transferred physical estimates retain the full correction and its norm-sum estimates, including all its masks.

6.1 The general-field local factor

Write $\vartheta = \bar{\Lambda}_K \bar{G}\psi$. At a good prime of norm Q put

$$V = Q^{-6z}, \quad W = \chi_u(p)Q^{-w}, \quad D = \psi(p)\bar{\chi}_u(p)Q^{-s}, \quad E = \vartheta(p)^6 Q^{4-6s}V.$$

The exact local sum is [2, (7.19)],

$$P_p = \frac{1}{1-V} + \frac{W}{1-W} + \frac{E}{1-E} \left(\frac{1-Q^{-1}}{1-V} + W \right) - \frac{D}{1-E} \left(1 + \frac{(Q-1)WV}{1-V} \right).$$

Thus the full factorization is

$$\zeta_K^S(6z) \frac{L_K^S(w, \chi_u)}{L_K^S(s, \psi\bar{\chi}_u)} H_{u,\psi}(s, w, z).$$

In particular the sixth-power row variable does not carry ψ^6 ; both principal poles remain at $w = 1$ and $z = 1/6$. Define $H_\psi^S(s) = H_{1,\psi}(s, 1, 1/6)$, and let $H_{\psi,1,Z}$ denote the principal-row correction after all compensations and prime-window sums have been performed.

6.2 The positive normalizer

Use the original geometry

$$\ell = \frac{1}{6}, \quad b = \frac{1}{8}, \quad M = \frac{5}{6}, \quad l_x = \frac{17}{48}, \quad l_y = \frac{23}{48}, \quad h = \frac{13}{16}, \quad C(s) = s - \frac{11}{16}. \quad (6.4)$$

Here $X = Z^{l_x}$ and $Y = Z^{l_y}$. Choose a sufficiently fine fixed even slot count J_0 , with $\ell_i = \ell/J_0$, scales $P_i = Z^{\ell_i}$, and nonnegative tests W_i on disjoint annuli. These analytic tests, lengths, and the count precede the later target in $\mathcal{F}_\omega$. After fixing that target and its arithmetic data, choose the finite identity-ray filter $T = T_\psi$ needed by the local presentation, and let $\mathcal{P}_i$ be the corresponding primes in the i th annulus. Put

$$S_i(Z) = \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{-5/6}, \quad A_T(Z) = (-1)^{J_0} Z^{-\ell/6} \prod_i S_i(Z). \quad (6.5)$$

The fixed-ray prime theorem gives $S_i(Z) > 0$ eventually and $A_T(Z) \asymp_{K,T,W_i} (\log Z)^{-J_0} > 0$. The ray filter changes a fixed positive density and the lower threshold. It follows that $A_T(Z)^{-1} \ll_{\psi,\varepsilon} Z^\varepsilon$ for every preassigned $\varepsilon > 0$. No uniform ray-class density is required.

At the principal residue $u = 1, w = 1, z = 1/6$, Q's local estimate is still $B_p = -1 + O(q_p^{-7/8})$. Choose, before the target,

$$\mu_0 = \frac{7}{16} \min_i \ell_i = \frac{7}{96J_0} > 0.$$

Then uniformly in $\Im s$ on $\Re s > 7/8$,

$$H_{\psi,1,Z}(s, 1, 1/6) = H_\psi^S(s) Z^{\ell/6} A_T(Z) \{1 + O_\psi(Z^{-\mu_0})\}. \quad (6.6)$$

The same fixed set S is used throughout. Enlarging its fixed cutoff as in Q, (10.1)–(10.2), gives

$$H_\psi^S \text{ holomorphic on } \Re s > 7/8, \quad |H_\psi^S(s) - 1| \leq \frac{1}{2}. \quad (6.7)$$

On the principal residue $u = 1, w = 1, z = 1/6$, the displayed local formula gives, uniformly for $\sigma = \Re s \geq 7/8$,

$$H_p - 1 \ll Q^{-2} + Q^{-1-\sigma} + Q^{3-6\sigma}, \quad B_p + 1 \ll Q^{-\sigma} + Q^{-1} + Q^{3-5\sigma} \ll Q^{-7/8}.$$

The first bound is normally summable. A fixed enlargement of S makes the logarithmic Euler-product tail as small as desired and proves (6.7). Once this has been done the quotient $B_p = G_p/H_p$ is legitimate on this principal locus. For each fixed positive window, its relative error is at most $O(P_i^{-7/8})$, by division by the same positive sum $S_i(Z)$. Thus (6.6) holds with the pretarget $\mu_0 = 7/(96J_0)$, even if T was selected later.

We make the base probe normalization explicit, separately from the prime-slot tests W_i . Choose the Bruhat–Schwartz test $W_{0,\text{add}}$ compactly supported in $K_S^\times$ and the nonnegative nonzero annular test $W_{1,\text{add}}$, as in [2, (7.1)–(7.3)]. Put

$$\mathcal{M}_{\text{add}}(z) = \int_{K_S^\times} \widehat{W}_{0,\text{add}}(y) |y|_S^z d^\times y, \quad \widehat{W}_{1,\text{add}}(w) = \int_0^\infty W_{1,\text{add}}(t) t^w \frac{dt}{t}.$$

The first test is chosen with $\widehat{W}_{0,\text{add}}(0) = 0$ and $\mathcal{M}_{\text{add}}(1/6) \neq 0$. At one complex place, subtract suitable multiples of two bumps on different radial subannuli: their additive integrals cancel while their different radial moments do not. The local zeta functional equation gives the required nonzero Mellin pairing, exactly as in P, (7.1). A fixed scalar then makes $\mathcal{M}_{\text{add}}(1/6) > 0$. Only $W_{1,\text{add}}$ is required to be nonnegative.

In this normalization P, (8.13), gives the unmarked double-residue scalar

$$c_S = \frac{1}{6} (\text{Res}_{s=1} \zeta_K^S(s))^2 \mathcal{M}_{\text{add}}(1/6) \widehat{W}_{1,\text{add}}(1) > 0.$$

The Gaussian normalization removes the target-dependent archimedean scalar before this computation. The physical principal row is unique: a row character trivial on all exterior primes is trivial by Chebotarev, so its Kummer class is a sixth power and its unit ambiguity belongs to $(R^\times)^6$. Any finer unit sectors used in positive moment estimates are recombined into this one physical orbit. Compensation commutes with unfolding and creates no additional principal residue.

With this calibrated positive scalar c_S , set

$$J_\psi(Z) = \frac{I_{\psi,\text{mod}}(Z)}{c_S A_T(Z)}, \quad f_\psi(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\psi^S(s)}{L_K^S(s, \psi)} ds. \quad (6.8)$$

Neither expression contains an analysis height cutoff. Lemma 3.4, the marked reflected energy in Appendix A, and the positive calculation of Q, Proposition 15.3 give, after the subpower reciprocal normalizer,

$$|J_\psi(Z)| \ll_{\psi,\varepsilon} Z^{3/16+\varepsilon}, \quad C(7/8) = 3/16. \quad (6.9)$$

No parameter optimization below 7/8 is needed.

7 The dynamic endpoint and one family closure

Fix (K, ω) and suppose $B = B_{K, \omega} > 7/8$. Proposition 4.2 gives $B \leq 11/12$, so (5.1) applies. Every companion denominator in the detector belongs to $\mathcal{F}_\omega$, hence its zero real parts are at most B . All common small-row and principal target contours use $B + e$. This is legitimate before any improved bound has been proved.

7.1 The capacity comparison

Use $\delta = 2a - 1$, $q = x\delta$, with $0 \leq \delta \leq \alpha = 5/6$ and $0 \leq x \leq 1/2$. At effective row width one, an inverse witness of length $r < 1$ has capacity $(1 - r)/2$. A plain witness of length $m < 1/2$, used twice in the fourth moment, has the actual capacity

$$z_P(m) = \frac{1 - 2m}{6\kappa} = \frac{1 - 2m}{9/2 + 12\Delta}. \quad (7.1)$$

The baseline $2(1 - 2m)/9$ is used only for comparison.

For the same witness and spike choices as Q, Proposition 19.2, the difference in the plain row-count exponent is

$$2q(1 - 2m) \left(\frac{2}{9} - \frac{1}{9/2 + 12\Delta} \right) = \frac{24q(1 - 2m)\Delta}{(9/2)(9/2 + 12\Delta)} \leq \frac{\Delta}{4} + O(\varepsilon). \quad (7.2)$$

The comparison is used where $m \geq 1/3 - O(\varepsilon)$ and $2q \leq 1$. Zero-capacity endpoints use the independent-scale inverse amplification or the zero-slot plain moment, rather than a strict marked estimate at zero width.

Put

$$\begin{aligned} D_x &= 3 - \frac{17x}{9}, & P_x &= \left(2 - \frac{8x}{9} \right) (1 - x), \\ J &= (\alpha - \delta)D_x + \delta P_x, & R_* &= 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2J}. \end{aligned}$$

The crossing of the short and long witnesses in Q, (19.7) and (20.8), now gives the actual bound

$$R \leq R_* + \frac{\Delta}{4} + \varepsilon. \quad (7.3)$$

This uses only the inverse and paired plain inputs above. The witness/row conversion is Q, Sections 8 and 19 applied with the transferred moments and fixed-type Hecke continuation. The conductor powers are unchanged; fixed local gamma parameters affect only height constants. One has $35/54 \leq J \leq 5/2$ on the whole closed parameter rectangle.

The only finite exceptional presentations among the sixth-power-free physical rows lie in the finite fixed sector part. Indeed, a prime outside S with valuation from one to five ramifies the sextic character; a character with fixed conductor in S has no such prime. Class representatives and U_S/U_S^6 give finitely many remaining sectors. This assertion concerns physical rows; canonical sixth-power descendants retain the exceptional treatment in P.

7.2 The original numerical certificate

If $U = Z^d$ is a moderate row norm, the full correction in T gives the relative exponent

$$\mathcal{E}(d; R, q) = -\frac{1}{48} + \frac{2}{3}\delta + \frac{q}{6} - h(1 - R) + (d - h) \left(R + \frac{\delta}{2} - \frac{17}{50} \right), \quad h = \frac{13}{16}, \quad (7.4)$$

measured against $C(7/8) = 3/16$. The term $q/6$ is already a base- Z exponent. Changing R by c changes $\mathcal{E}$ by dc . Write $E_*(d) = \mathcal{E}(d; R_*, q)$.

The exact certificate Q, (20.9), is unchanged. Set $y = 1/2 - x$ and $v = 51 + 41y$. Then

$$10368vJ(-E_*(h)) = (3 + 5y)\{(4v\delta - 79)^2 + 49\} \\ + 4y\{4v\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y\}. \quad (7.5)$$

Every displayed summand is nonnegative for $0 \leq y \leq 1/2$ and $0 \leq \delta \leq 5/6$. Since $v \leq 17(3 + 5y)$ and $J \leq 5/2$,

$$-E_*(h) \geq \frac{49}{440640} > 10^{-4}. \quad (7.6)$$

The crucial comparison for the actual dynamic capacity is therefore

$$\mathcal{E}(h; R, q) - \{C(B) - C(7/8)\} \leq -\frac{49}{440640} + \frac{h}{4}\Delta - \Delta + O(\varepsilon) = -\frac{49}{440640} - \frac{51}{64}\Delta + O(\varepsilon). \quad (7.7)$$

Thus the loss in capacity is absorbed by the very gap that defines the contradiction. No fourth moment with $\kappa < 3/4$ is used.

For the bounding exponent $R_* + \Delta/4$ the coefficient of $d - h$ in (7.4) is at least $33/50 - \delta/2 \geq 73/300$. Hence the endpoint controls the selected moderate ranges below h . The supply margin is

$$\frac{\ell}{h} = \frac{8}{39} > \frac{7}{37}, \quad \frac{8}{39} - \frac{7}{37} = \frac{23}{1443}.$$

A fixed extension $h < d \leq h + \zeta$, $\zeta < 1/48$, leaves available length greater than $1/5 > 7/37$. At $\delta = 5/6$ the unmarked amplified endpoint gives $R = 1 - \delta$; no shrinking strict inverse width is used.

7.3 The remaining ranges

Take $d_{\min} = 1/100$. The source comparisons retained in T give the following positive margins before small losses.

Range	Saving
Floor bin $\delta = 1/50$, before subtracting Δ	7/1200
$d_{\min} \leq d \leq 1/2$, no selected slots, before subtracting Δ	49/14400
Small rows $1 \leq U \leq Z^{d_{\min}}$, apart from the principal row, relative to $C(B)$	63/800
Principal w contour	23/960
Principal z contour	13/9600
Principal slot approximation	$\mu_0 = 7/(96J_0)$

For the middle range, the unselected count $R = 76/75 - 2\delta/3$ yields

$$\mathcal{E}(d; R, q) \leq -\frac{529}{2400} + \frac{25}{96}\delta \leq -\frac{49}{14400}.$$

For the small rows the full-correction estimate on $(\text{Re } s, \text{Re } w, \text{Re } z) = (B + e, 1/2, 17/50)$ gives $-79/800 + 2d_{\min} = -63/800$, relative to $C(B)$. This includes the nontrivial unit sectors. There is no additional cost from using $11/12$ instead of an individual target boundary: the common line is $B + e$ throughout. Rows above $h + \zeta$ are estimated on $(2, 2, z_{\infty})$; the arbitrary-decay assertion in T allows a fixed sufficiently large z_{∞} .

7.4 Real choices before the target

The following is Q, Section 20.6's order of choices applied to the family $\mathcal{F}_\omega$. Put

$$m_{\text{hi}} = \frac{51}{64}\Delta, \quad m_{\text{high}} = \min\{m_{\text{hi}}, 63/800\}.$$

Reserve fixed fractions of the low margin Δ , the high margin m_{high} , and both principal contour margins. Make the choices common for the permitted representatives of the fixed finite collection of types $0, \omega_\infty, \bar{\omega}_\infty$. Choose the moment losses and capacity decrements first. The uniform real-parameter assertion of P then gives a slot mesh on $3/4 \leq \kappa \leq 5/6$. Choose an even J_0 sufficiently large that $2\ell/J_0$ is below that mesh, the allocated rounding loss, and $1/185$.

Take $\ell_i = \ell/J_0$ and choose nonnegative nonzero tests on disjoint intervals

$$I_i = \left(1 + \frac{2i-1}{2J_0+1}, 1 + \frac{2i}{2J_0+1}\right), \quad 1 \leq i \leq J_0.$$

These analytic supports are disjoint before any ray or row mask and are chosen before the finite-order target twist. The fixed arithmetic ray filter may be selected after that twist; it changes only constants and thresholds. The pretarget error exponent is $\mu_0 = 7/(96J_0)$ in (6.6). Now choose the amplitude and bin widths, remaining real losses, ζ , and z_∞ , in this order, reserving the fractions used in Q, Section 20.6. The real-parameter uniformity for these choices follows from the transfer and finite-depth argument in Appendix A.

For definiteness, after these allocations one can retain

$$\sigma = \frac{1}{8} \min \left\{ \frac{51}{64}\Delta, \frac{63}{800}, \frac{23}{960}, \frac{13}{9600}, \frac{7}{96J_0} \right\} > 0. \quad (7.8)$$

This number depends on the fixed family and Δ , but not on its later target ψ . Notice that $\sigma < \Delta/2$.

7.5 Height choices and continuation

Now fix a target $\psi \in \mathcal{F}_\omega$, removing its global norm twist. Choose between it and its conjugate to satisfy the nonzero angular condition in Section 3.3. Conjugation preserves the family and the real parts of zeros, so proving the estimates for this representative suffices. Choose its arithmetic data, final S , finite sectors, and identity-ray filter T . Then choose all retained internal seminorm and Sobolev orders. The transferred estimates give finite numbers A_ψ, B_ψ and a positive permitted height ceiling such that, for the subsequently chosen external order N ,

$$|J_\psi - f_\psi| \ll_{\psi, N} Z^{C(B)-2\sigma} (1 + T_1)^{A_\psi} + Z^{B_\psi} T_1^{-N}.$$

Degree, infinity type, and fixed local profiles can enlarge A_ψ, B_ψ , and the threshold. Choose a positive τ_ψ small enough for the height ceiling and the reserved saving, put $T_1 = Z^{\tau_\psi}$, and then choose N large enough. With the strict reserves above, the conclusion of Q, Lemma 11.1 is

$$|J_\psi(Z) - f_\psi(Z)| \ll_\psi Z^{C(B)-\sigma}. \quad (7.9)$$

The analysis height belongs only to this estimate: neither J_ψ nor f_ψ depends on T_1 . The low bound with loss $\Delta/2$ and $\sigma < \Delta/2$ now gives

$$|f_\psi(Z)| \ll_\psi Z^{C(B)-\sigma}. \quad (7.10)$$

The continuation is elementary once this common exponent is available. Moving the defining line of f_ψ arbitrarily far right gives rapid decay as $Z \downarrow 0$. Hence

$$\mathcal{F}_\psi(s) = \int_0^\infty f_\psi(Z) Z^{-C(s)} \frac{dZ}{Z}$$

is holomorphic on $\operatorname{Re} s > B - \sigma$. Mellin inversion, first on $\operatorname{Re} s = 2$, identifies it with

$$e^{(s-5/6)^2} \frac{H_\psi^S(s)}{L_K^S(s, \psi)}.$$

Both numerator factors are nonzero on $\operatorname{Re} s > 7/8$, and $B - \sigma > 7/8$. Thus every $L_K(s, \psi)$ in $\mathcal{F}_\omega$ has no zero in $\operatorname{Re} s > B - \sigma$. At a norm-character pole its reciprocal merely has a zero, which causes no problem.

By the definition of B , if $B > 7/8$ there exists a target in $\mathcal{F}_\omega$ with a zero of real part greater than $B - \sigma/2$. The preceding conclusion is a contradiction. Consequently $B \leq 7/8$, and in particular $L_K(s, \omega)$ is zero-free in $\operatorname{Re} s > 7/8$. Apply this with $\omega = \psi \circ N_{K/F}$ and use Lemma 2.1. This proves Theorem 1.2.

8 Original sources and proof dependencies

The main theorem concerns arbitrary number fields and arbitrary fixed unitary infinity types. Its analytic source dependence is Source Input 1.1. The general-field conclusions **T**, **I**, and **P** are the proved transfer of Proposition 3.1, not further hypotheses.

Source or argument	Role in this paper
P, Sections 3, 5–7 and Appendix A	Original theta construction, arithmetic calibration, and local coefficient identities; its general-field and fixed-type realizations are verified in Section 3.
Q, Sections 5–16	Original physical and marked-reflection constructions; norm-form Poisson, additive Gram, and full local compensation are verified here.
Q, Sections 17–18	Original long canonical proofs. Appendix A supplies the general-field replacements, keeping their integer width and capacity calculations.
Q, Sections 19–20	Witness comparison and the exact endpoint certificate; the dynamic capacity loss and fixed-type family closure are calculated in Section 7.
Abelian descent and the fixed-type starting bound	Proved here, including norm-character poles and exact sixth-power amplification masks.

Q’s published finite-order $7/8$ theorem and P’s cyclotomic $1 - 10^{-6}$ theorem are not used as a shortcut to an arbitrary-field conclusion. In particular $K = F(\zeta_{12})$ need not be cyclotomic over $\mathbb{Q}$. The transfer is the reason the original constructions apply to this K .

The higher-order large sieve of [4, Theorem 1.3] is compatible with the exterior ideal symbols, but it does not alone prove the canonical inverse moment. At column length $N = M^{9/10}$ its term $(MN)^{2/3}$ is $M^{19/15}$, exceeding the desired $M^{1+\varepsilon}$. The two source Poisson reductions and the marked terminal are therefore essential. Likewise, the actual identity $\chi_n(a^6) = \mathbf{1}_{(n,a)=1}$ is retained throughout; replacing it by a primitive character and discarding its zeros would invalidate the amplification.

This paper reproduces the transfer arguments and the simplified closure. It does not independently re-prove every line of the long original theta and canonical constructions. That is the remaining source-relative qualification in Theorem 1.2.

A Transfer of the canonical estimates

Proposition A.1 (Fixed-field moment transfer). *Retain the original canonical arguments of Q, Sections 14, 17 and 18, and the normalized theta realization of P, with its general-field and fixed-infinity-type realization established above. Then the directed marked inverse estimate (4.3) and the paired fourth moment (5.3) hold over the fixed field K with their displayed width, coefficient, zero-extension and common-mask restrictions. For the fourth moment one may use $3/4 \leq \kappa \leq 1$; when $\kappa < 1$, its prime premise concerns only finite-order characters and is the boundary $(1 + \kappa)/2$. The slot mesh is independent of the subsequently fixed slot count. For a fixed infinity-type orbit, the real losses and that mesh can also be chosen before the later finite-order target twist and its fixed conductor. Thus the two general-field moment estimates are consequences of the cited canonical proofs and the replacements verified here, rather than additional moment hypotheses.*

Proof.

1. The common orbital Poisson formula. Use Lemma 3.4, including its general-periodic version and separate zero orbits. The primitive character formula is used in the two inverse transforms and their initialization; the full finite Fourier formula is used in the second plain transform. The mask is expanded first. Its common norm kernel, fixed logarithmic growth, and finite-seminorm bounds are exactly the profile data required by the source reductions.

2. Local tables, masks, and fixed unitary coefficients. At a good prime of norm P the full normalized finite sums obey

$$|G(\mathfrak{p}^a, h)| = P^{(a-1)/2} \mathbf{1}_{v_{\mathfrak{p}}(h)=a-1} \quad (6 \nmid a), \quad G(\mathfrak{p}^a, h) = P^{a/2} \mathbf{1}_{\mathfrak{p}^a|h} - P^{a/2-1} \mathbf{1}_{\mathfrak{p}^{a-1}|h} \quad (6 \mid a).$$

These follow by separating the higher residue digits and applying finite-field orthogonality. CRT gives the full two-column correlations, including collision zeros, exactly as in Q, Section 13.3. Thus a displayed sixth-divisible exponent remains a puncture. All ideal common-support extractions are exact before any coefficient is bounded or a coprime support is enlarged. Once factor ideals and the finitely many unit sectors are fixed, the old-label fibre is fixed modulo Γ ; this replaces the source's uniqueness of primary generators, at a fixed finite cost. The fixed background stays on columns, not in an exterior row modulus. Lemma 4.1 gives its exact propagation: the first and second transforms replace ν only by $\nu\vartheta$, $\vartheta \in \Theta$, using $\nu(ACn) = \nu(A)\nu(C)\nu(n)$ and $|\nu(C)|^2 = 1$. No finite Fourier expansion of a genuine ν is made. Character-equivariant theta periods retain the fixed unit phases; their conductor phases are multiplicative row, tuple, or frozen scalars. They are not dropped by replacing the period by an unweighted unit average. Cube completion leaves $\nu(b)^3$ on the extracted cube variable, as already checked in that lemma.

3. The full marked terminal. The identity $\mathbf{1}_{p|A} = 1 - \chi_p(A)^0$ applied to the compatible theta branches gives inactive and active mark sizes P^{-1} and $P^{-1/2}$. A residual row digit $j = 1$ and a mark digit $j = 0$ have trivial pair phase. The remaining row character is quadratic and the active marked squarefree column is cubic. Freeze every other digit, including $j = 4$, and separate fixed local sectors of the row product and marked product individually. The annular theta kernel established above, the fixed-field quadratic sieve verified in [2, Lemma 6.4], and the cubic sieve of [4, Theorem 1.3] therefore give Q, Lemma 14.3's full reflected energy. In its notation this exponent is

$$E_{\text{ref}} = \frac{O}{2} + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u - \ell_b - \frac{1}{2}(T_d - v - 3\ell_b)_+, \quad u = \min\{v, z_a, (v + z_a)/3\}.$$

Here O is the powerful-row length, H the actual residual squarefree-row length, v, ℓ_b the dual squarefree and cube lengths, and S_0, B_0, T_d are the local extraction quantities of that lemma. Its

completed quadratic reduction, Q, Lemma 5.5, is an ideal factorization with all collision masks retained. No special low-side condition $2A_0 \leq O$ is used here. For the squarefree fourth-power label of length V , puncture length Q , row length M , and completed scale N_* , the primewise conductor comparison is

$$H \leq M - O + o(1), \quad T_d - S_0 - B_0 \leq 2M - O + Q + V - N_* + 2z_a + o(1).$$

These are Q, (17.11), with ideal norms. If $F_0 = N + V$ and

$$F_0 - M - Q - z_0 \geq c, \quad 4F_0 - 3M - 6z_0 \geq c,$$

the short completion $V < d_0$, $N_* = N - 3b_h$, $b_h < d_0 + o(1)$ gives the three terminal bounds

$$F_0 - c, \quad F_0 - 2c + 5d_0, \quad F_0 - M/4 - 3c/4 + 5d_0/2,$$

apart from the assigned small losses. They are strictly below F_0 when $d_0 \leq c/200$, precisely Q, (17.12)–(17.14). The label mass is counted once and cancels its original Z^{-V} normalization. Neither a label prime nor a puncture prime has been incorporated into the fixed data.

4. The two inverse transforms and initialization. Use the orbital identity in paragraph 1 at Q, (17.22) and (17.36), retaining the genuine coprime expression before truncating its ratio tails. Its prefactor, conductor norm, puncture divisor, finite transform and scalar kernel argument are exactly the quantities in Q, (17.23)–(17.60). Paragraph 2 gives the same finite column identities and whole-product priority assignments. In particular the canonical coefficient remains $\bar{\Lambda}\gamma_2\nu\rho$, since the calibrated squarefree identities are $\gamma_2^3 = \mu_K\Lambda$ and $\bar{\Lambda}\gamma_2\gamma_1 = \mu_K G$. The two invariants in paragraph 3 and the strict child-row decrease in Q, (17.17), are unchanged. The common smooth kernel and all outside row masks remain common through the weighted Cauchy inequalities of Q, Section 17.3; no row-dependent column coefficient is introduced. The one-transform initialization has the same local identities. Its two margins, before enclosure losses, are

$$m - r - 2z + 2G, \quad 3m - 2r - 8z + 10G + 2P_1,$$

as in Q, (17.85)–(17.86). Here $G, P_1 \geq 0$ are the source's extracted norm lengths. The hypotheses of (4.3) supply the required positive margins. This imports the finite canonical induction and proves the directed inverse estimate, with its exact mask and coefficient restrictions. Proposition 4.2 gives the independent-scale form with the exact sixth-power masks.

5. The ordinary fourth-moment reflection and coefficients. The compatibility checks in Section 5 prove the fixed-infinity-type reflection and the paired propagation, so we use them rather than treating an infinite-order twist as a finite Fourier coefficient. The finite conductor factor remains $Q^{1/2-s}$ and the dual norm scale Q/X . The product of complex gamma quotients may have repeated poles at real part zero; its small annular coefficients are $O(2^{-j/2}(1+j)^A)$, which are summable. The degree changes orders and constants, not the conductor exponent. The common-mask identities Q, (18.4)–(18.8), are applied to the displayed presentation before primitive reduction. Apply the general-periodic version of paragraph 1 in the two plain transforms. CRT and complete factor allocation preserve Q, Lemma 18.2's one whole-product character on each separated side, the original individual slot coefficients, one common mask and norm power on both rectangles, and their equal product of plain scales. The two sides differ by a character in Θ . Thus they have the same finite exceptional status, and the actual prime characters remain finite order. Their prime estimate is Q, (18.12), with the stated finite-order zero-free premise. All strict width and positive-part ledgers in Q, Sections 18.4–18.8, consequently keep their coefficients.

6. Exceptional rows and the prime pool. The unmasked Mellin argument in Section 5, followed only afterwards by common-mask inclusion–exclusion, replaces the exceptional lattice estimate. If the four plain scales are at least Z^a , $a \geq 0$, and have common product T_0 , its equal-product cancellation gives $O(Z^\varepsilon T_0^{1/2} Z^{-a} (1 + |t|)^A)$ after central normalization, exactly the saving in Q, (18.47); a genuine background has zero residue and gives a stronger bound. The canonical exceptional condition prescribes each exterior row valuation modulo six. Hence its ideal is $\mathfrak{h}_0 \mathfrak{d}^6$ in finitely many unit sectors. At a newly nonunit extracted prime of common multiplicity one, the fixed factor has valuation two, so the row valuation is four modulo six. If these primes have length v_1 , then $N\mathfrak{h}_0 \geq Z^{4v_1}$, giving the count required by Q, (18.41). The old moving support is disjoint from this support because its original column zeros are still present. The comparison of h with hp^6 uses a fixed prime pool of length ℓ_{pool} ; the prime ideal theorem supplies $Z^{\ell_{\text{pool}} + o(1)}$ allowed primes. An output ideal has only a bounded number of such prime divisors. The same single pool-density gain therefore survives on unit orbits. The local error valuations 1, 6, 7 have squared coefficients $P^{-1}, 1, P^{-1}$ by paragraph 2. A mesh below $\ell_{\text{pool}}/2$ separates the pool from all live slots. These are the field-sensitive ingredients in the plain induction, and prove (5.3) by the cited induction.

7. Uniform real choices and finite propagation. The source chooses its width floor, strict decrements and finite depth from bounded real length ranges and requested losses. None of the replacements above changes a norm exponent or an affine ledger. A fixed K, S, ψ can add finite sectors, kernel logarithmic powers, seminorm orders and finite constants. For every preassigned $\eta > 0$, each such logarithm is bounded by $C_{K,S,\psi,\eta} Z^\eta$; the same η works whatever its fixed logarithmic degree. Allocate this aggregate loss through the source’s finite depth before choosing the internal orders. Support-enclosure constants affect only the eventual threshold through $O_{K,S,\psi}(1/\log Z)$. The width step fixes the prime-pool length, and then the mesh, before the number of slots is fixed. Divisor allocations for that fixed number of slots cost their one small-power loss; their constants, not the mesh, may depend on the slot count. It follows that real choices can precede the later finite-order twist and enlargement of S . Their constants and height orders need not do so. The common Fourier measures in paragraphs 1 and 4 allow the finite-seminorm propagation of Q, Section 17.3 and the loss order of Section 18.8 unchanged. Increasing a final external tail order differentiates only external tests, not an already chosen internal moment profile, so its height degree does not increase. Finer unit sectors used in positive moment estimates are recombined into the original $(R^\times)^6$ physical orbits; they add no principal residue. The analytic annuli, slot lengths, and number of slots precede the target conductor. The fixed arithmetic ray filter can be chosen afterwards: it changes the positive normalizer by a fixed density, while the relative local slot error and its preassigned exponent μ_0 are unchanged. This proves the asserted quantitative and mesh uniformity and completes the transfer. $\square$

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